# EECE 5554 Lab 5: Inertial Odometry

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## Overview

This report presents the inertial odometry analysis for circle and square walking datasets using IMU data. The figures follow the required order and illustrate sensor behavior, integration results, and reconstructed trajectories.

## 1 Circle Dataset

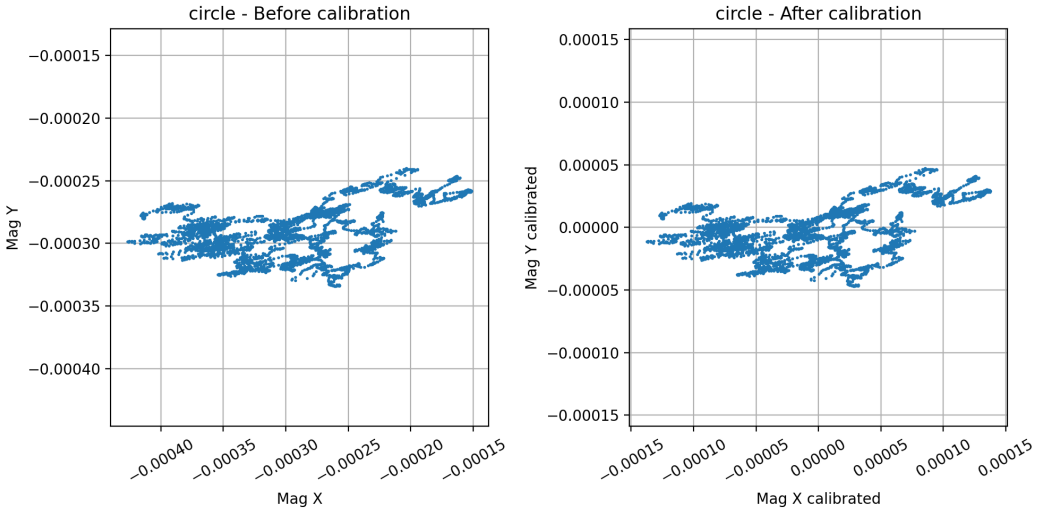



Figure 1: Magnetometer calibration (before vs after).

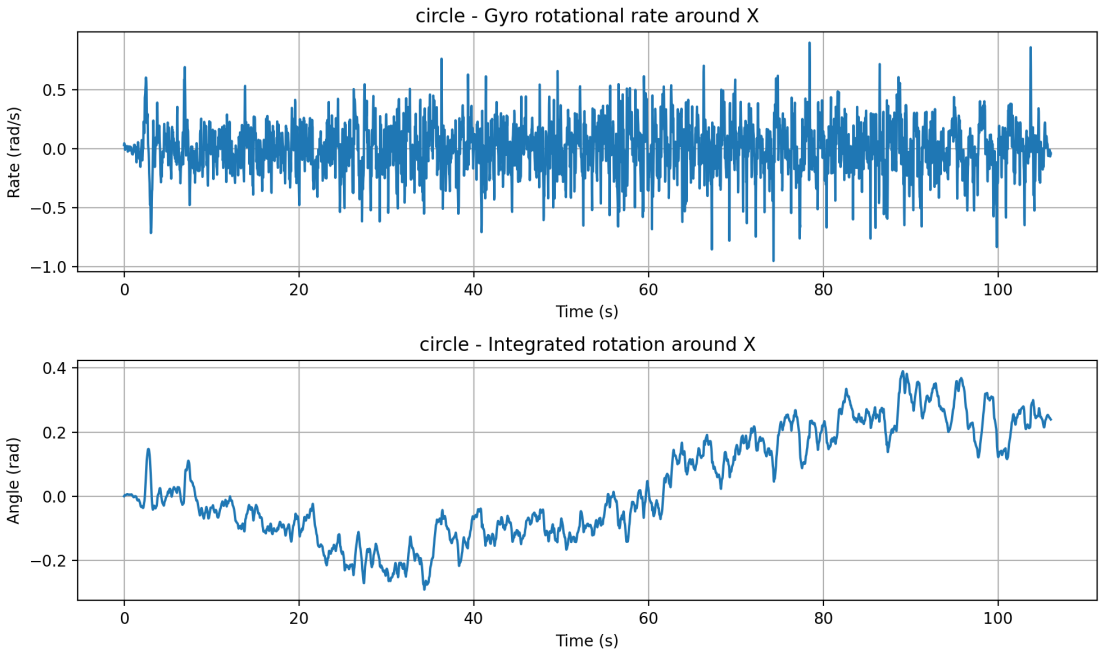



Figure 2: Gyro X: rotational rate and integrated rotation.

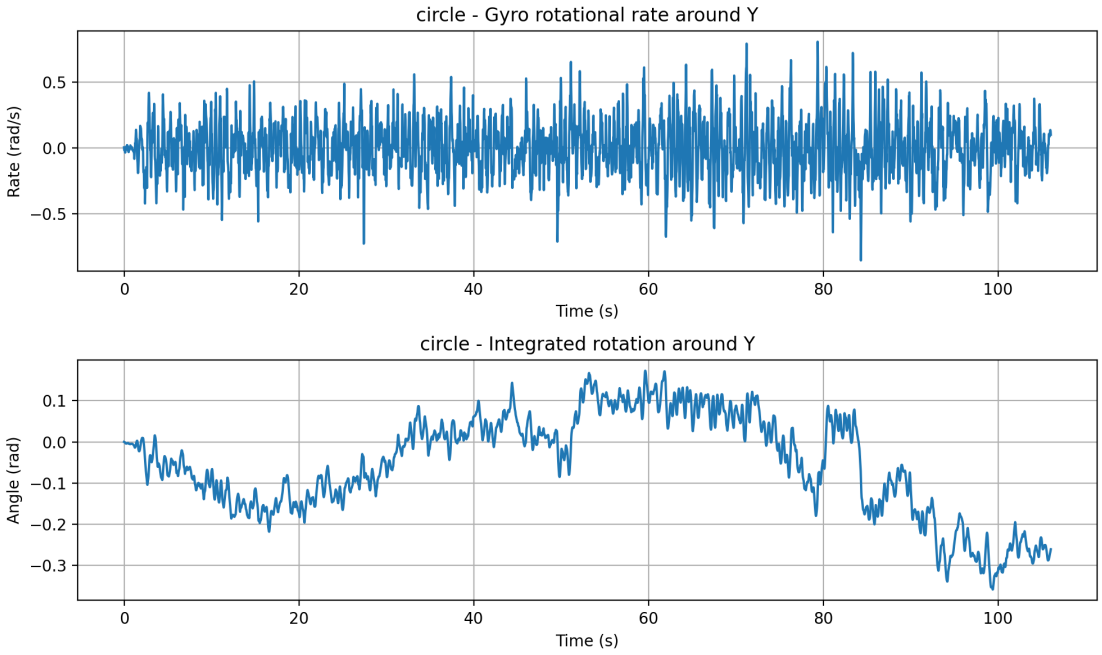



Figure 3: Gyro Y: rotational rate and integrated rotation.

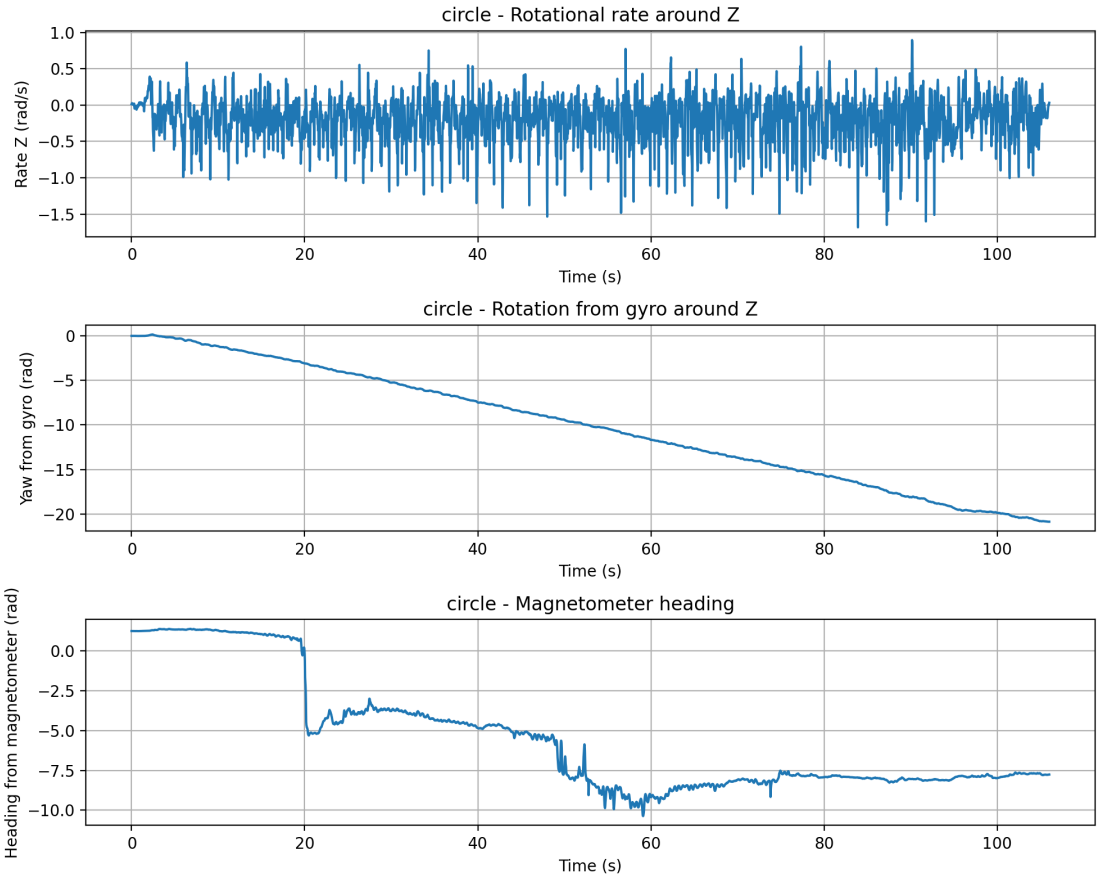



Figure 4: Gyro Z, integrated rotation, and magnetometer heading.

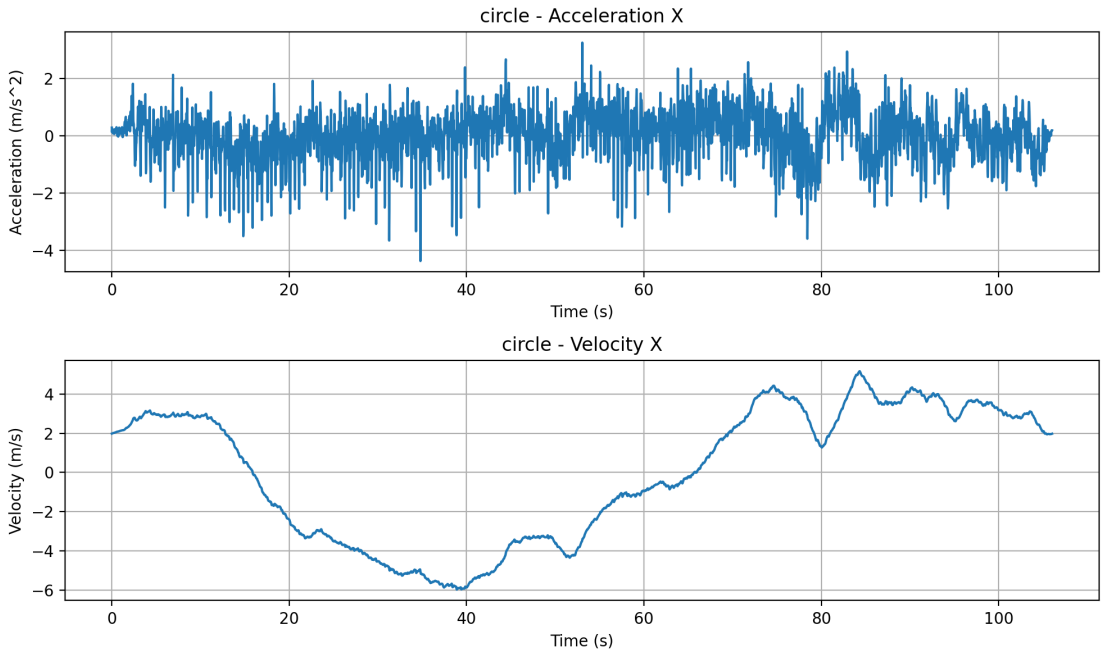



Figure 5: Acceleration and velocity (X axis).

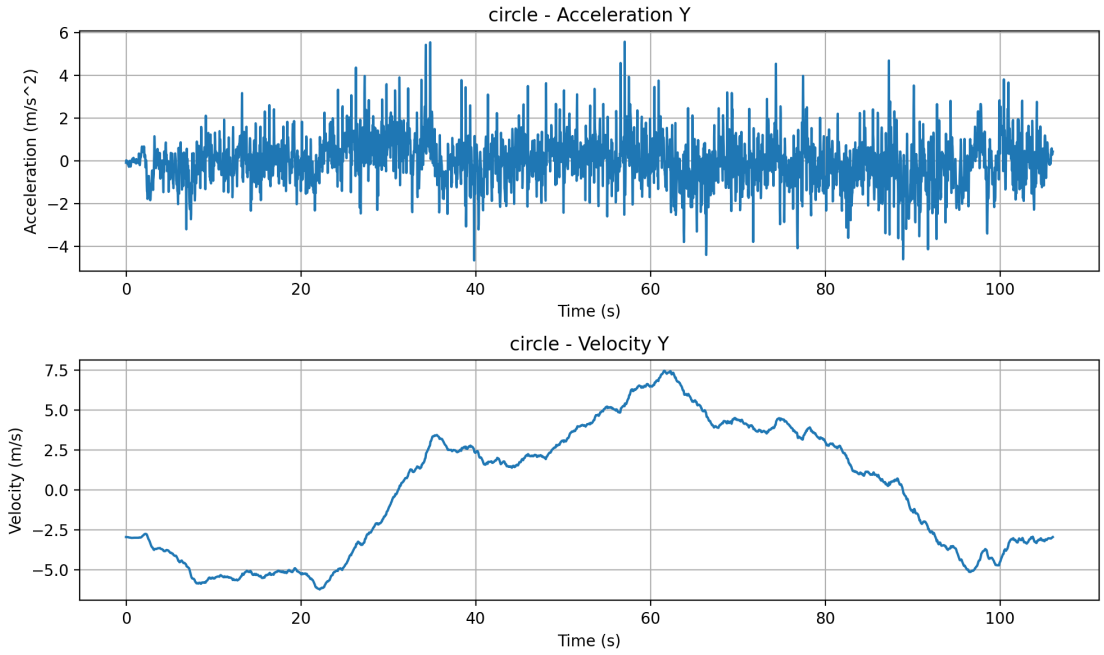



Figure 6: Acceleration and velocity (Y axis).

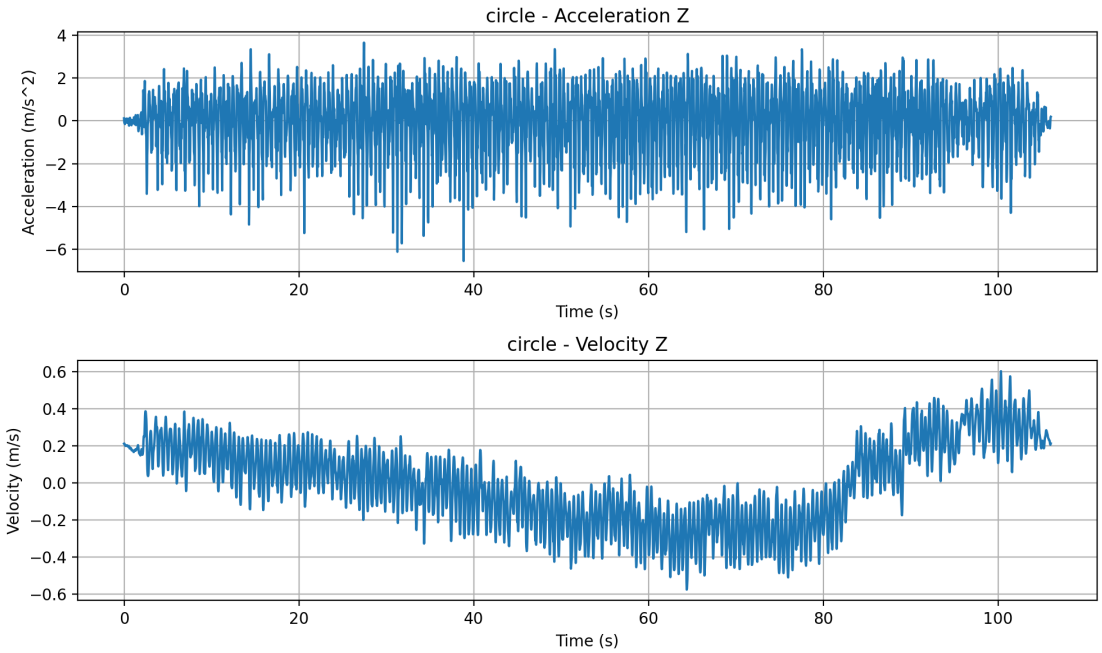



Figure 7: Acceleration and velocity (Z axis).

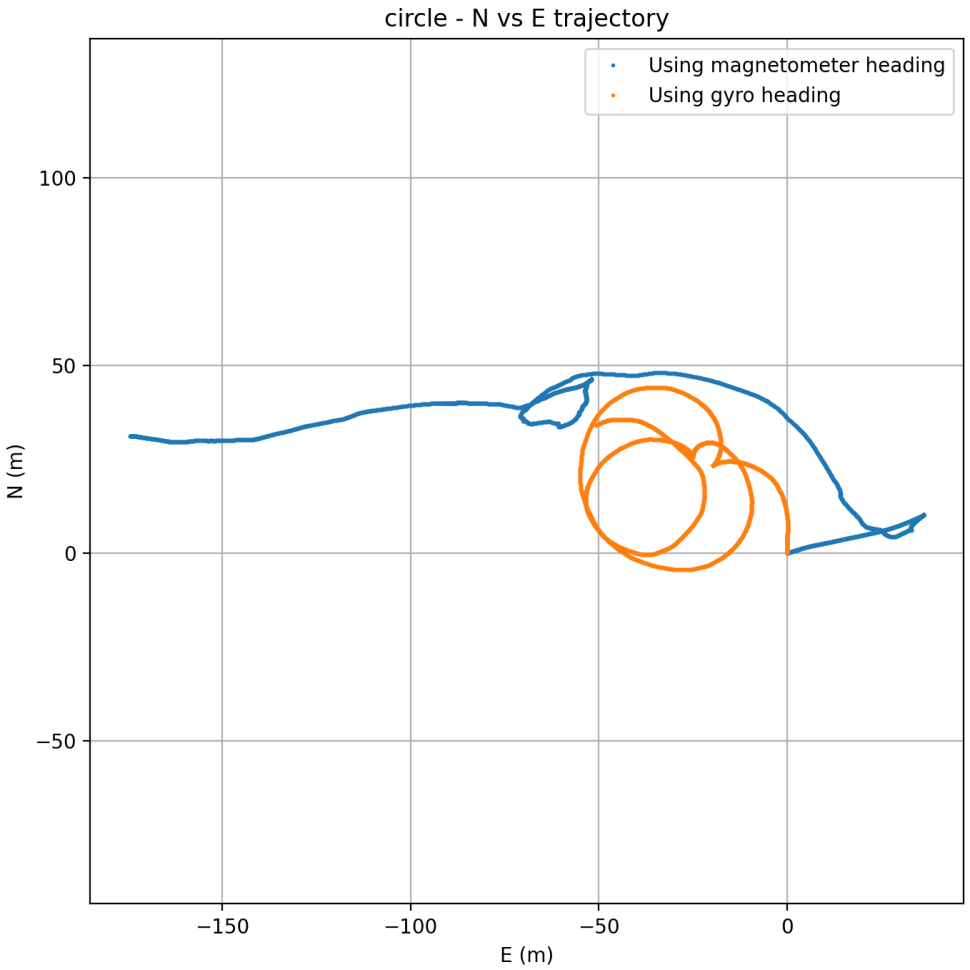



Figure 8: Estimated N vs E trajectory.

## 2 Square Dataset

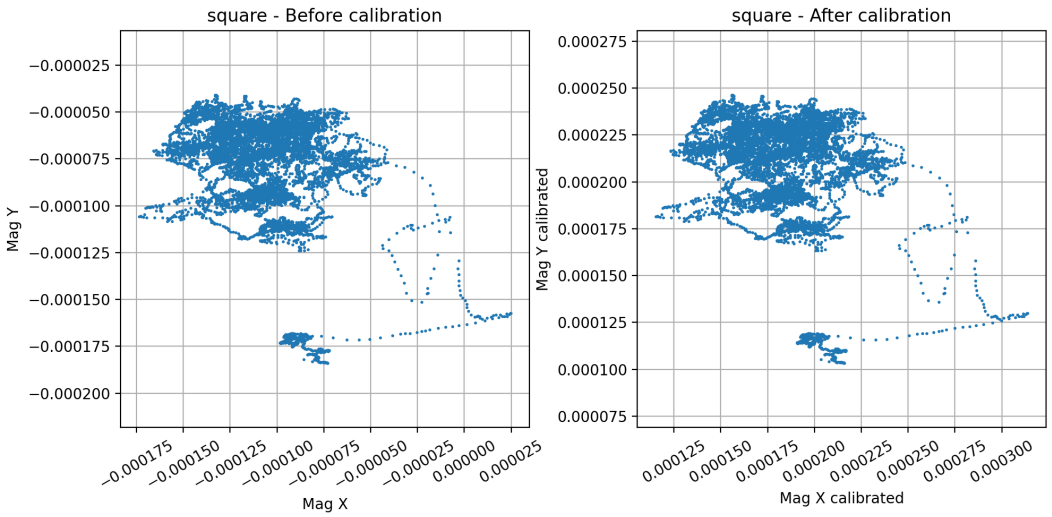



Figure 9: Magnetometer calibration (before vs after).

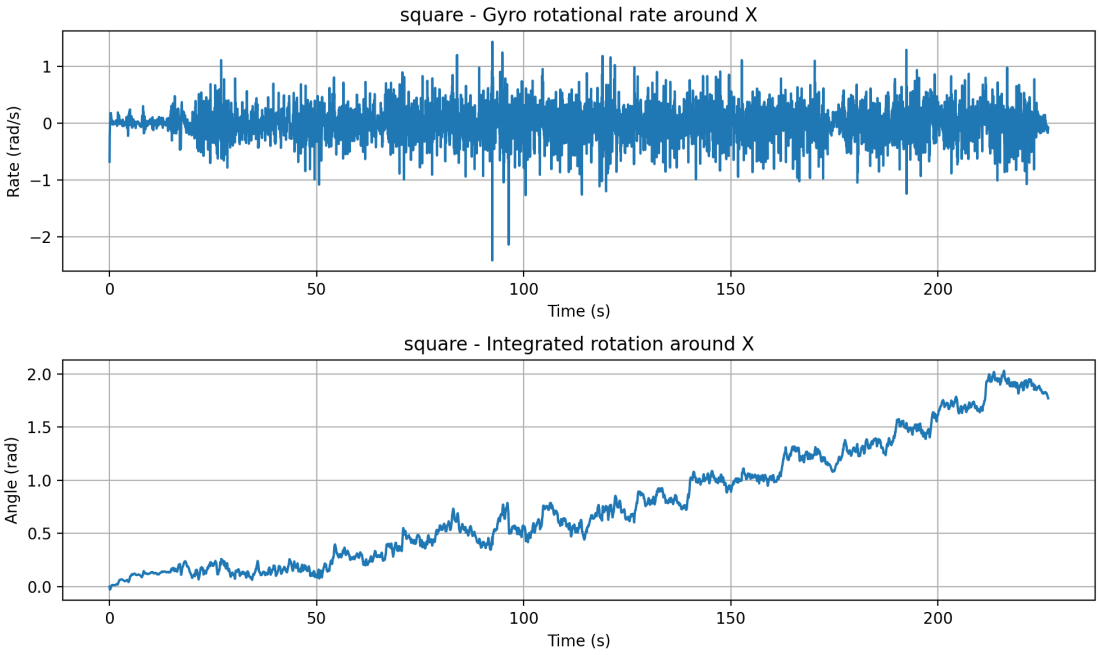



Figure 10: Gyro X: rotational rate and integrated rotation.

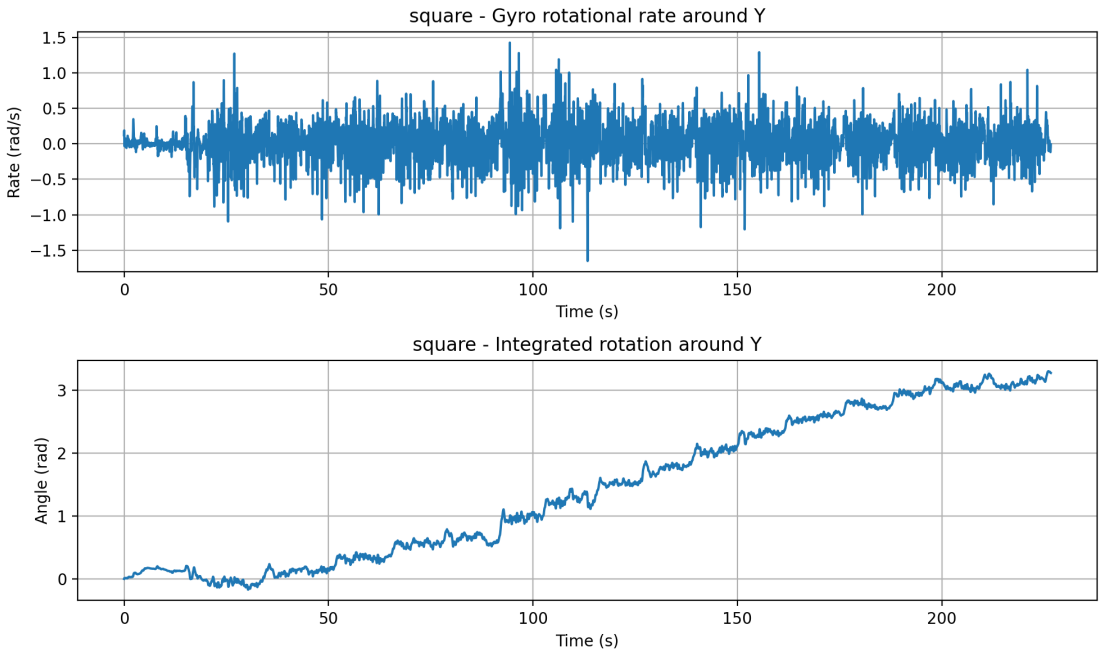



Figure 11: Gyro Y: rotational rate and integrated rotation.

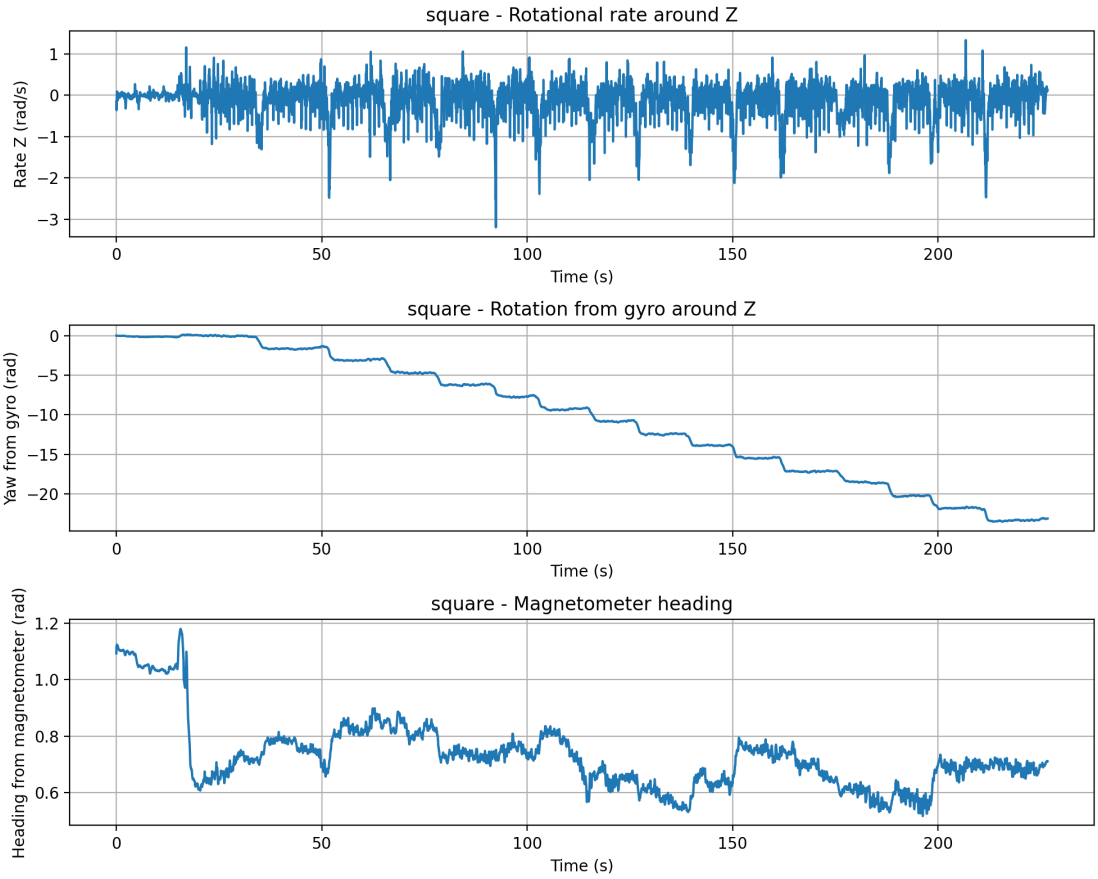



Figure 12: Gyro Z, integrated rotation, and heading.

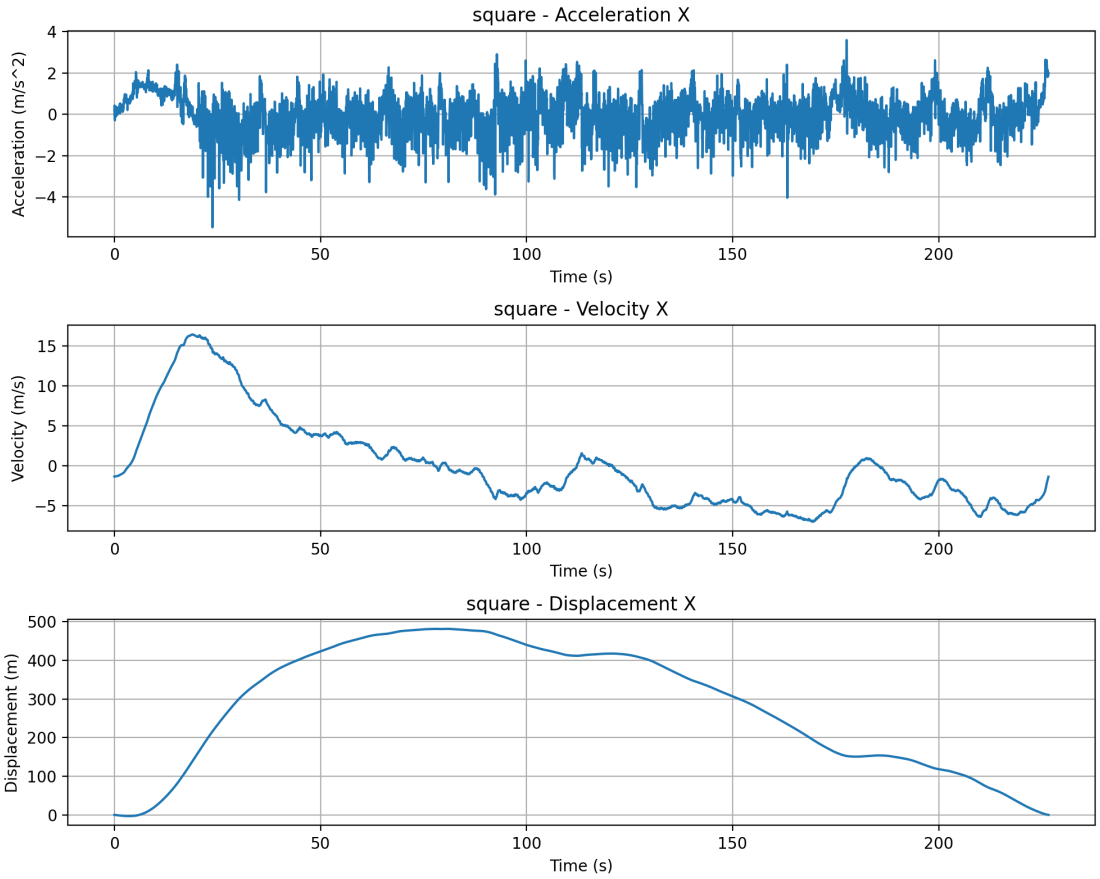



Figure 13: Acceleration, velocity, and displacement (X axis). Displacement is obtained via double integration for additional analysis.

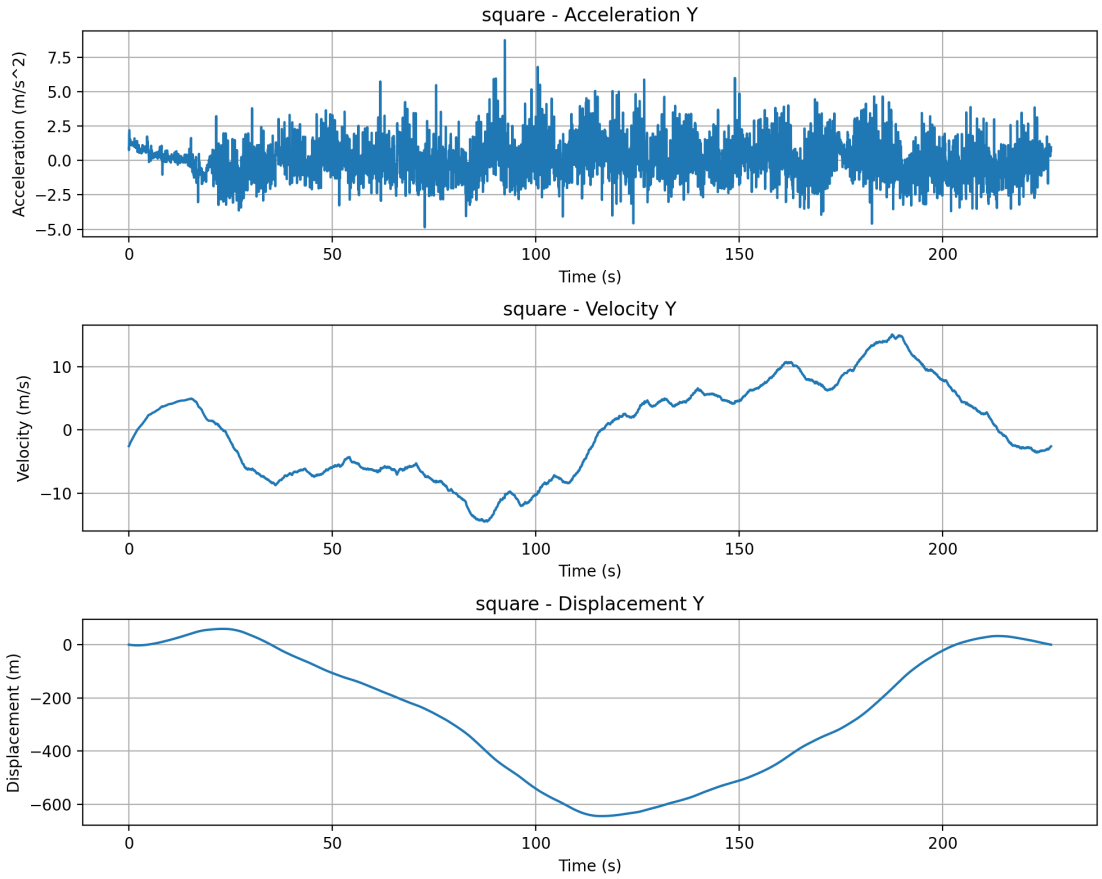



Figure 14: Acceleration, velocity, and displacement (Y axis). Displacement is obtained via double integration for additional analysis.

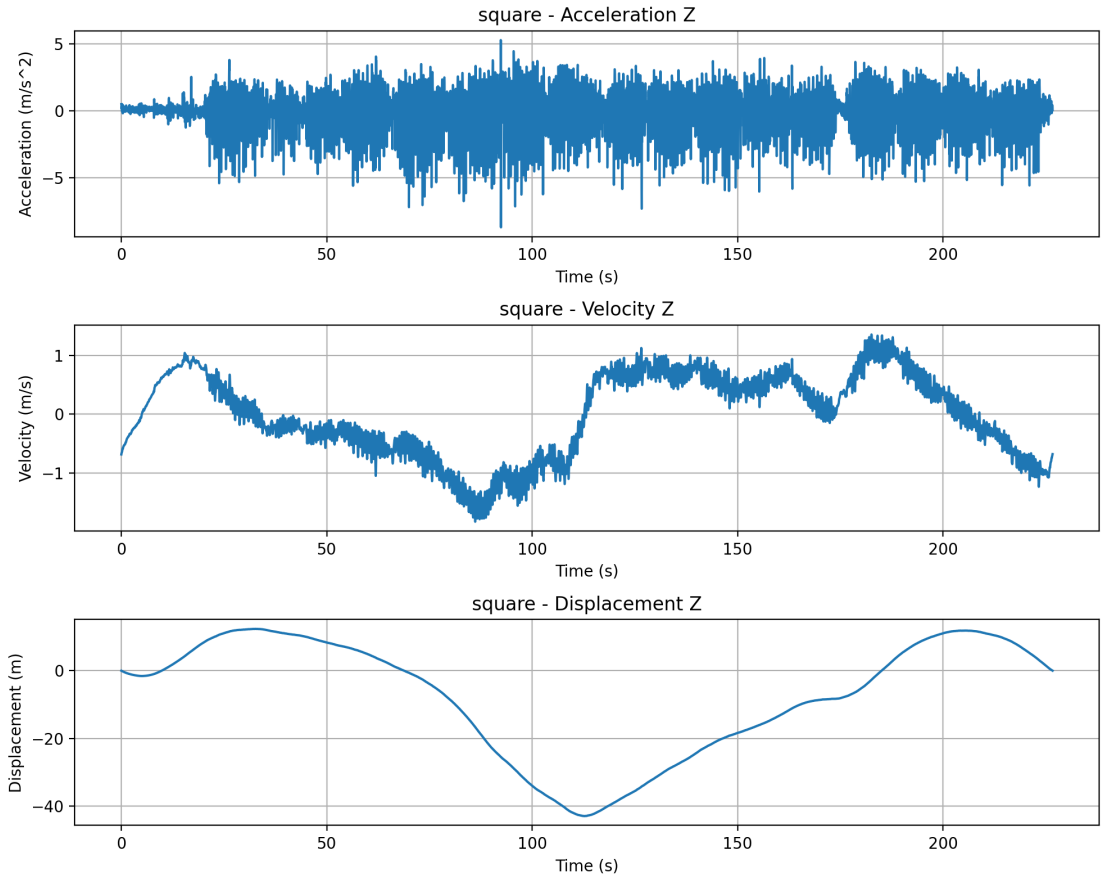



Figure 15: Acceleration, velocity, and displacement (Z axis). Displacement is obtained via double integration for additional analysis.

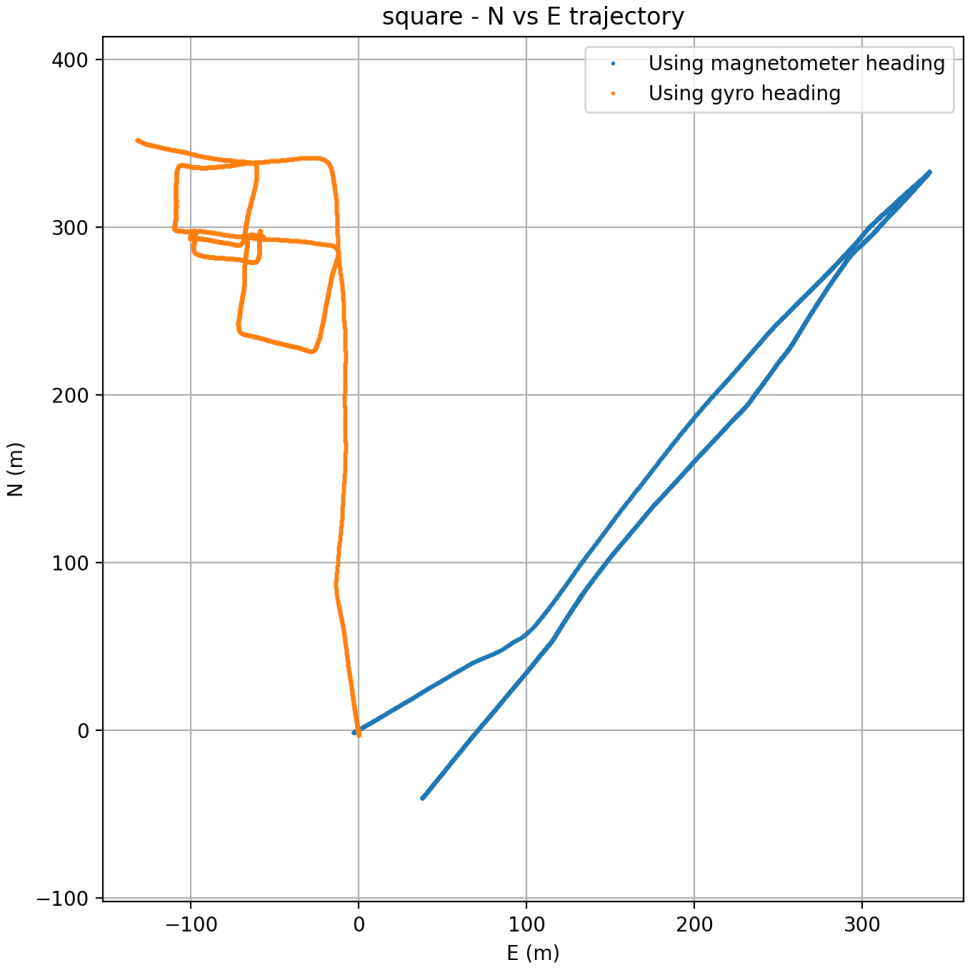



Figure 16: Estimated N vs E trajectory.

# 3 Discussion

### Do the results make sense?

The reconstructed trajectories generally follow the expected circle and square motion patterns, but there is noticeable drift and distortion over time. For example, the trajectory does not perfectly close and deviates by several meters after one loop, indicating accumulated integration error. This behavior is expected because inertial odometry relies on integrating noisy IMU signals, which amplifies small errors in acceleration and orientation into large position deviations. Compared to the ground truth video, the overall shape is roughly preserved, but the path is not geometrically accurate.

### How to improve data collection?

To improve data quality, the IMU should be mounted rigidly and aligned consistently with the motion direction instead of being hand-held. Reducing vibration, maintaining a steady walking pace, and ensuring proper calibration before data collection would also help. For future vehicle experiments, using a stable mounting platform and minimizing external disturbances will significantly reduce integration drift.